

IoT & ML-based Vehicle Speed Monitoring with RFID-Enabled License Validation

Project ID: 25-26J-314



The Research Problem & Its Gravity

Current Traffic Enforcement Problems in Sri Lanka

✦ Data Gap

- Speed cameras identify vehicles, not drivers
- No real-time driver identity verification

✦ Reaction Gap

- Drivers receive fines weeks later
- No opportunity for immediate behavior correction

✦ Safety Gap

- No system monitors driver fatigue/emotion + speed together

Stakeholders Affected & Benefited

Drivers

- No real-time feedback on risky behavior
- High accident exposure

Traffic Authorities

- Manual & reactive enforcement
- Limited real-time visibility

Emergency Services

- Delayed accident response
- No predictive risk insights

Government & Society

- High accident-related losses
- Public safety concerns

Existing Technologies

- Weather-dependent systems
- Vehicle-only identification

Design Excellence of the Overall Solution

- End-to-End Integrated Architecture
- Dual Identity Verification
- Real-Time Feedback Loop
- All-Weather Reliability
- Scalable & Modular Design





IoT System for RFID-Based Vehicle Speed Monitoring System

Specialization in Information Technology



IoT System for RFID-Based Vehicle Speed Monitoring System

- Introduces real-time vehicle entry detection at highway access points
- Uses ESP32 and RC522 RFID readers for IoT-based data collection
- Identifies both driver (human) and vehicle using dual RFID tags
- Calculates vehicle speed using fixed RFID checkpoints
- Stores real-time monitoring data in a centralized database



IoT System for RFID-Based Vehicle Speed Monitoring System

User Requirements Addressed

- Real-time vehicle entry and speed monitoring
- Identification of driver and vehicle together
- Automated speed calculation without manual enforcement
- Reliable data storage for monitoring and analysis

Contribution

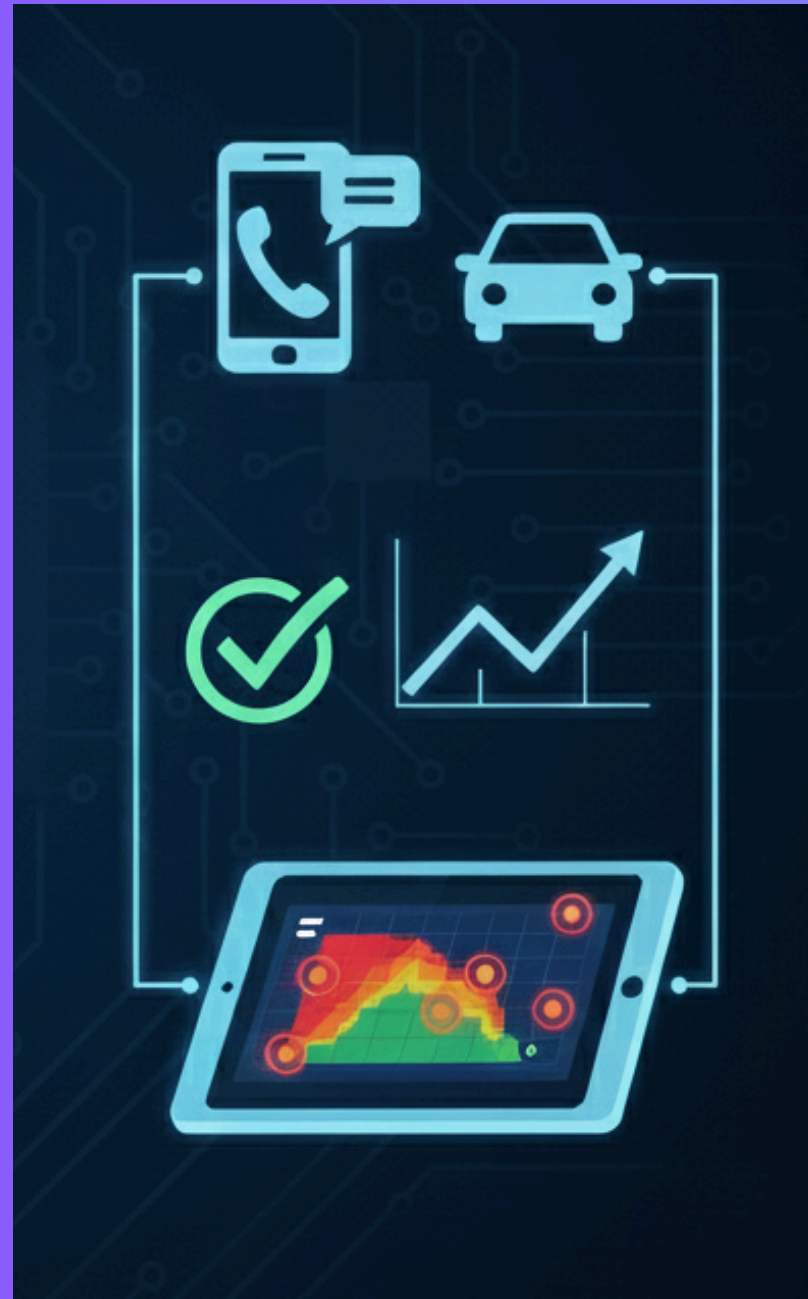
- Introduces RFID-based live speed monitoring to Sri Lanka
- Reduces dependency on police speed gun enforcement
- Links driver identity with speed behavior
- Provides accurate IoT-generated data for intelligent traffic systems

Dashboard & Driver Alert System

Specialization in Information
Technology



Dashboard & Driver Alert System



- Eliminates delayed enforcement through instant SMS/Email alerts
- Converts speed violations into real-time driver awareness
- Enables immediate behavioral correction during the same trip
- Provides live violation visibility to authorities

Dashboard & Driver Alert System

User Requirements Addressed

- ✓ Real-time monitoring of speed violations
- ✓ Secure access for traffic authorities
- ✓ Instant notification to drivers
- ✓ Clear and simple violation history view

Contribution

- ✓ Introduces real-time behavioral feedback in traffic enforcement
- ✓ Enables measurement of driver behavior change after alerts
- ✓ Bridges and IOT and ML outputs into a single enforcement interface
- ✓ Supports data-driven traffic policy and decision making

Intelligent Accident Response and Statistical Resource Prioritization Component

Specialization in Information
Technology



Intelligent Accident Response and Statistical Resource Prioritization Component

- Enables automatic accident detection using onboard sensors
- Performs AI-based driver injury assessment in real time
- Forecasts near-future high-risk zones using statistical models
- Supports both emergency response and preventive deployment



Intelligent Accident Response & Statistical Resource Prioritization

User Requirements Addressed

- ✓ Rapid identification of accident severity
Immediate emergency notification for critical cases
- ✓ Data-driven prioritization of patrol
patrol and medical resources
- ✓ Clear risk insights for
traffic authorities

Contribution

- ✓ Introduces a unified accident response and forecasting component
- ✓ Bridges reactive emergency handling with proactive planning
- ✓ Enables data-driven resource allocation for authorities
- ✓ First integrated ML + statistics-based traffic safety solution proposed for Sri Lanka

Driver State Accident Prevent System

Specialization in Information
Technology



Driver State Accident Prevent System



- Developed a driver-state accident prevention component to reduce accident likelihood by monitoring driver condition
- Identifies drowsiness and driver's emotion
- Triggers an alert when risk is high to regain driver focus

Driver State Accident Prevent System

User Requirements Addressed

- ✓ Real-time drowsiness detection
- ✓ Real-time emotion recognition
- ✓ Risk level output (High / Low) from combined results
- ✓ Immediate intervention (adaptive music)

Contribution

- ✓ Combines drowsiness and emotion for better risk estimation
- ✓ Adds context-aware music intervention
- ✓ Provides clear real-time driver state feedback
 - ✓ Supports accident prevention by improving focus & mood

IoT & ML-based Vehicle Speed Monitoring with RFID-Enabled License Validation

Customer Persona

- ✓ Primary users are national traffic enforcement authorities

Market Size

- ✓ Sri Lanka's expressway network with scalability to smart cities

What Is Unique in Our Solution

- ✓ Real-time, driver-identified enforcement instead of delayed vehicle-based fines

Commercialization & Sustainability

- ✓ Government-to-Government smart traffic platform

Cost Recovery Model

Annual licensing and analytics services

Repository Link

<https://github.com/vrbbro18/Research-Project.git>

Thank You

